

Chapter Two

Information Technology(IT)

Dr. Farah Qais Abdullah

HARDWARE AND SOFTWARE

To understand the myriad uses of IT in many fields needs to be familiarized with computer terminology, hardware, and software applications. Every computer performs similar functions. Specific hardware is associated with each function.

Input devices take data that humans understand and digitize those data, that is, translate them into binary forms of ones and zeroes, on and offs that the computer processes; a **processing unit** manipulates data; **output devices** produce information that people understand; memory and **secondary storage devices** hold information, data, and programs (Figure below).



Components of the computer system.

-Components of a computer system

1- Hardware

- The term hardware refers to the physical components of your computer such as the system unit, mouse, keyboard, monitor etc.

2- Software

- The software is the collection of instructions that make the computer work. For instance, when you type in words via the keyboard, the

software is responsible for displaying the correct letters, in the correct place on the screen. The software is holding either on your computer's hard disk, CDROM, DVD or on a diskette (floppy disk) and is loaded from the disk into the computers RAM (Random Access Memory), as and when required.

3- **The Human Being:** The brains behind the whole system

Hardware

The physical components of a computer are called **hardware**. Pieces of the hardware are categorized according to the functions of each performs input, process, output, and storage. As you recall, inside the computer, all data are represented by the **binary digits (bits)** 1 (one) and 0 (zero). To translate data into 1s and 0s is to **digitize**.

Input Devices

Input devices function to take data that people understand and translate those data into a form that the computer can process.

The Input devices are dividing into two categories: **keyboards** and **Direct-entry devices**.

Direct-entry devices include pointing devices, scanning devices, smart and optical cards, speech and vision input, touch screens, sensors, and human-biology input devices.

A Mouse is one of the input devices, which used in conjunction with the GUI (Graphical User Interface), point and click. Other types of Input Devices: Trackballs, Light Pens, Touch Screens, Tablets

Output devices

Once data are processed, output devices translate the language of bits into a form humans can understand. Output devices are divided into two basic categories: those that produce a **hard copy, including printers and plotters**; and those that produce **soft (digital) copy, including monitors** (the most commonly used output device). Soft copy is also produced by speakers that produce speech, sound, or music.

Peripheral device

A peripheral device is any device, which you can attach to your computer. Thus, you could attach a scanner or modem to the back of your system unit.

Input and output devices

Input devices

Input devices

- Keyboard
- Scanner
- Touchpad
- mouse
- trackball²
- joystick
- microphone
- camera (web, digital)

Output devices

- monitor
- projector¹
- printer
- plotter
- speakers
- earphones

Projector is a device used to project a computer image or other images from independent devices, such as DVD players, Blu-ray player, etc. onto canvas or a wall.

Trackball, unlike a mouse, is not movable. Hand movements are transmitted to the screen by rolling the ball which is located on the upperside of the device.

Input and output devices are

1- **Storage devices**, due to the necessity for writing and reading data, they are classified as input/output devices.

2- **Touchscreen** (i.e. monitor sensitive to touch) is out device while displaying computer image, and at the same time input device while receiving manual orders.

(<https://en.wikipedia.org/wiki/Touchscreen>)

Common devices and media for storing and transferring data:

- Hard disk
- CD and DVD discs
- USB flash drive
- Diskette and ZIP diskette
- Memory cards
- Internal hard disk
- external hard disk
- Network disk drive
- Online storage device

(https://en.wikipedia.org/wiki/Computer_data_storage)

➤ **SOFTWARE**

The software is, unlike hardware, intangible part of the computer. It consists of a sequence of commands, written according to strict rules. Programmers, in various programming languages, write programs.

Software types:

1- **The operating system** is a program, which manages computer hardware. First computers did not have operating systems; they had programs that were directly loaded into the computer (e.g. punch

cards). Today, computers have an operating system that loads into the computer's memory during its startup. Computer functions are based on its operating system. Within the operating system, drivers (responsible for the functioning of a computer) and various utility programs (responsible for the functionality of a computer) are installed.

There are a number of different types of operating system in common use. The IBM PC (Personal Computer) was introduced way back in 1981 and was originally supplied with an operating system called DOS (Disk Operating System). This operating system was very basic, and you had to be a bit of a computer expert just to understand how to use it. It was NOT user-friendly. Later on, Microsoft introduced Windows and this is the operating system that is most widely used on PCs today. To complicate matters further, there are a number of different types of Windows.

The first widely used version of Windows was called Windows 3.1. This was more powerful than DOS and far easier to use. It had a Graphical User Interface (GUI), i.e. you could 'drive' it using a mouse and drop down menus.

Later versions of Windows include Windows 95, Windows NT, Windows 98, Windows 2000 and Windows XP. The good news is that later versions of Microsoft Windows look almost identical to each other and are all much easier to use than Windows 3.1.

وهناك أنواع مختلفة من نظم التشغيل المستخدمة حالياً. لقد ظهر أول حاسوب شخصي في عام 1981 وكان من إنتاج شركة آي بي إم (IBM) وكان يستخدم نظام تشغيل يسمى "دوس". وقد كان هذا النظام بسيطاً (غير متطور)، وكان يتعين أن تكون لديك خبرة في عالم الحاسوب لتعرف كيف تستخدمه. فلم يكن "سهل الاستخدام". وبعد ذلك، قدمت شركة مايكروسوفت نظام "ويندوز" الذي أصبح الآن النظام الأكثر شيوعاً مع أجهزة الحاسوب. هناك إصدارات مختلفة من نظام ويندوز. لقد كان ويندوز 3.1 هو أول إصدار لويندوز يتم استخدامه على نطاق واسع. وقد كان أكثر فعالية من نظام دوس وأسهل كثيراً في الاستخدام. فهو يستخدم واجهة تطبيق رسومية (Graphical User Interface - GUI)، أي أنه يمكنك التعامل معه من خلال الفأرة والقوائم المنسدلة. وظهر بعد ذلك ويندوز 95 و98 وNT و2000. والجدير بالذكر أن الإصدارات الأخيرة من ويندوز تقريبا كلها متشابهة وكلها أسهل في الاستخدام مقارنة بويندوز 3.1.

The most famous operating systems are:

A- Operating systems from Microsoft:

Dos, Windows 3.11, Windows 95, Windows 98, Windows Millennium, Windows 2000, Windows XP (professional, Home), Windows 2003 server, windows vista, windows seven.

B- Operating systems from other companies:

Linux, Unix, Macintosh.

What does the operating system do?

1. Control the hardware and software of your computer.
2. Sets up the look and feel of your computer interface.
3. Enable all other programs to run.
4. Instruct the screen to display data entered with the keyboard.

1- Applications Software (البرامج التطبيقية)

What is an application program?

- An application

program is the type of program that you use once the operating system has been loaded.

Examples include word-processing programs (for producing letters, memos etc), spreadsheets (for doing accounts and working with numbers), databases (for organizing large amounts of information) and graphics programs (for producing pictures, advertisements, manuals etc).

A. Word processing (معالج الكلمات)

A word processing program (such as Microsoft Word) allows you to easily produce letters, memos, etc. You can easily mail merge a list of names and addresses to produce mass mailers, individually addressed to customers or subscribers.

Microsoft Word

<http://www.microsoft.com/office/word/default.htm>

WordPerfect <http://www.corel.com/Office2000>

- Produce a catalog containing graphics.
- Creating letters, faxes, memos, and business documents.

B. Spreadsheets (الجدول الالكترونية)

- A spreadsheet program (such as Microsoft Excel) allows you to work out a company's income, expenditure and then calculate the balance. It enables you to make 'what if' type projections of how the company will fair in the future and to forecast how changes in prices will affect profits.

Microsoft Excel <http://www.microsoft.com/office/excel>

Summary

- 1- Makes financial projections. انشاء مشروعات مالية
- 2- Produce charts and graphs. انشاء رسومات بيانية
- 2- Budget account and deal with them حساب الميزانية والتعامل معها

C. **Databases** قواعد البيانات

A database program (such as Microsoft Access) allows you to compile information and then to search this information to extract just the information you require.

Microsoft Access <http://www.microsoft.com/office/access>

Summary

- 1- Sorting a large amount of information.
- 2- Creating a list of contact details

D. **Presentation** (العروض التقديمي)

A presentation program (such as Microsoft PowerPoint) allows you to produce professional-looking presentations, which can be printed out directly onto slides for use with an overhead projector. Alternatively, you can display your presentations directly on a computer screen.

Summary

A presentation application creates overhead projections or slides

Microsoft PowerPoint

<http://www.microsoft.com/office/powerpoint>

E. Accounts / Payroll (انظمة الرواتب)

In most large organizations, the accounts are maintained by a computerized system. Due to the repetitive nature of accounts, a computer system is ideally suited to this task and accuracy is guaranteed.

في معظم المنظمات الكبيرة، يتم الاحتفاظ بالحسابات من خلال نظام محوسب. ونظرا لطبيعة المتكررة للحسابات، هي مناسبة لنظام الكمبيوتر بشكل مثالي لهذه المهمة، ودقة المضمون

Summary

The accounting application controls payroll and stock control.

يقوم برنامج انظمة الرواتب بحساب رواتب الموظفين والتحكم في المخزون

F. Web browsing (متصفحات الانترنت)

These applications allow you to view and interact with the World Wide Web (WWW).

Microsoft Internet Explorer <http://www.microsoft.com>

G. Web authoring (انشاء مواقع الويب)

These applications allow almost anyone to create a website, quickly and easily.

Microsoft FrontPage: <http://www.microsoft.com>.

H. A multimedia application Creates interactive learning and training materials.

برنامج الوسائط المتعددة يقوم بانشاء مواد تعليمية وتدريبية

I. **Desktop publishing** produces a newspaper and magazines with complex page layouts.

برنامج النشر المكتبي يقوم بإنشاء صحف ومجلات ذات تخطيطات معقدة

What is systems development?

This is a rather general term used to describe the way new software is specified, computer coded, tested and then delivered to the user.

هو عبارة عن مصطلح عام يصف إنشاء برنامج جديد واختباره حتى يصل إلى المستخدم

What is a systems development cycle?

ما المقصود بمراحل تطوير النظم

Most IT projects work in cycles. First, the needs of computer users must be analyzed. This task is often performed by a professional Systems Analysts who will ask the users exactly what they would like the system to do, and then draw up plans on how this can be implemented on a real, computer-based, system.

يتم تطوير أغلب مشروعات تكنولوجيا المعلومات على مراحل. أولاً، يتم تحليل احتياجات مستخدمي الحاسوب. وعادة ما يقوم بذلك المهمة محللو نظم محترفون حيث يستفسرون عن رؤية المستخدمين للنظام والمهام التي سيقوم بها ثم يقومون بوضع الخطط المتعلقة بكيفية تنفيذ ذلك من خلال نظام فعلي يعتمد على الحاسوب.

The programmer will take the specifications from the Systems Analyst and then convert the broad lines into actual computer programs. Ideally, at this point there should be testing and input from the users so that what the programmers produce is actually what they asked for.

ويقوم المبرمج بعد ذلك بأخذ المواصفات من محلل النظم ويحول هذه الخطوط العريضة إلى برامج حاسوب فعلية. ويجب أن تجرى بعد ذلك اختبارات لهذه البرامج وتعرض على المستخدمين للتأكد من أن النظام حسب طلبهم.

Finally, there is the implementation process during which all users are introduced to the new systems, which often involves an element of training.

Once the users start using the new system, they will often suggest new improvements and the whole process is started all over again. These are methodologies for defining a systems development cycle and often you will see four key stages, as listed below.

Analysis, Design, Programming, and Testing

يتم تطوير أغلب مشروعات تكنولوجيا المعلومات على مراحل. أولاً، يتم تحليل احتياجات مستخدمي الحاسوب. وعادة ما يقوم بتلك المهمة محللو نظم محترفون حيث يستفسرون عن رؤية المستخدمين للنظام والمهام التي سيقوم بها ثم يقومون بوضع الخطط المتعلقة بكيفية تنفيذ ذلك من خلال نظام فعلي يعتمد على الحاسوب.

ويقوم المبرمج بعد ذلك بأخذ المواصفات من محلل النظم ويحول هذه الخطوط العريضة إلى برامج حاسوب فعلية. ويجب أن تجرى بعد ذلك اختبارات لهذه البرامج وتعرض على المستخدمين للتأكد من أن النظام حسب طلبهم.

وأخيراً، تتم عملية التطبيق والتي يتعرف خلالها المستخدمون على النظم الجديدة، وعادة ما تتخللها فترة تدريب.

وبعد أن يبدأ المستخدمون في استخدام النظام الجديد، من الممكن أن يطالبوا بإضافة تعديلات عليه وتبدأ العملية من البداية ثانية.

هناك عدة طرق لتعريف مراحل تطوير النظم ولكن بشكل عام، ستجد المراحل الأربع الأساسية التالية: 1- دراسة الجدوى 2- التصميم 3- البرمجة 4- التطبيق

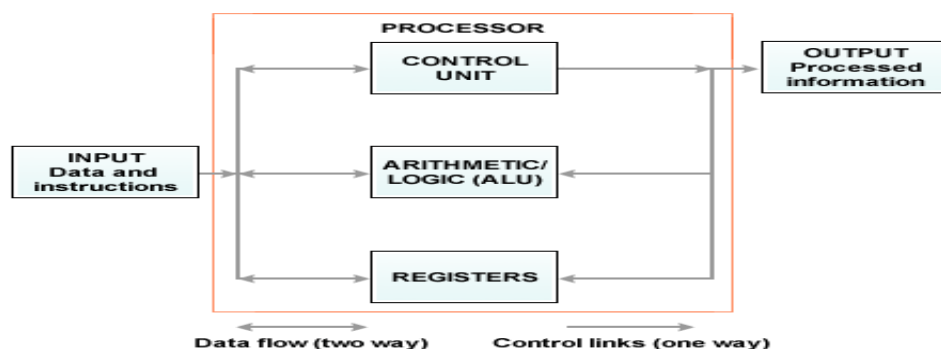
➤ HARDWARE

It contains from the following

- 1- **Processing devices(CPU)**
- 2- **Network devices**
- 3- **Memory (primary and secondary storage)**
- 4- **Input devices**
- 5- **Output devices**

1- **Processing devices (CPU)**

Such as the central processing unit. The CPU is the brains within your computer. It performs most of the calculations within the computer and is responsible for the smooth running of your operating system (Microsoft Windows) as well as your application programs, such as word-processors, spreadsheets, and databases. There is a small amount of memory associated with the CPU, which it uses to perform these operations. It also accesses and uses the main memory (RAM - Random Access Memory) within your computer. In many ways, the CPU is the single most important item within your computer that governs the overall speed of your computer.



The CPU's speed is measured in MHz. This relates to the frequency that the CPU runs at and the higher the MHz rating of your CPU the faster your computer will run. To give you some indication of how PCs have advanced over the years, the original IBM PC released in 1981 ran at less than 5 MHz while modern PCs can run at speeds well in excess of 2000 MHz. Note that 1000 MHz is the same as 1 GHz.

CPU process only 0s and 1s, all data are translated into computer languages into a series of these binary digits or bits.

The microprocessor has different portions which perform different functions:

- 1-**Control Unit:** this controls the flow of information.
- 2-**Arithmetic Logic Unit (ALU)** performs arithmetic calculations.
- 3-**Registers:** which store the very small amount of data and instructions for a short period of time.

Control unit

- Direct and coordinates all units of the computer to execute program steps.
- Direct and coordinate all operation of the computer systems.

These operations include;

- 1- Control to the input and output devices.
- 2- Entry and retrieval of information from memory.
- 3- Routing of information between the memory, arithmetic and logic unit.

Control unit automatically coordinates the operation of the entire computer system, although the control unit does not perform any actual processing on the data, it acts as a central nervous system uses to the sent control signal to other units.

Arithmetic and Logic Unit (ALU)

Perform the processing of data including arithmetic operation such as addition, subtraction, multiplication, division, and logic operation including comparison (ex. $A < B$) and sorting.

Registers

Registers are devices capable of storing information, receiving data from other areas within the computer and transferring information as directed by the control unit, it is used for temporary storage of data or instruction and the most important register are :

- 1- Program counter (PC): it contains the address of the next instruction to be executed.
- 2- Instruction register (IR): it contains the instruction being executed.
- 3- Address register (AR): holds the address of memory location.

2- **Network devices** such as Ethernet controllers, LAN, network interface cards(NIC) and cables.

3- **Storage devices (Includes the primary and secondary storage)**

A- Primary storage (main memory): The memory is the part of the computer that holds information (data and instruction) for processing so name because small amounts of data and information that will be immediately used by the CPU are stored there.

The specific functions of the main memory are to hold (store):

- 1- All data to be processed.
- 2- Intermediate results of processing.
- 3- Final results of processing

B- Secondary Storage: where the much larger amount of data and information (an entire software program, for example) are stored for an extended period of time. such as hard disk drive, compact disk(CD), digital versatile disk(DVD), floppy disk.

Memory Capacity

Bit: All computers work on a binary numbering system, i.e. they process data in ones or zeros. This 1 or 0 level of storage is called a bit. Often hardware is specified as a 32-bit computer, which means that the hardware can process 32 bits at a time. The software is also described as 16 bit, 32 bit or 64-bit software.

CPU process only 0s and 1s, all data are translated into computer languages into series of these binary digits or bits

Eight bits are needed to represent a character. This 8-bit string is known as a byte.

The storage capacity of a computer is measured in bytes. The hierarchy of byte memory capacity is as follows:

- a) Byte:** A byte consists of eight bits.
- b) Kilobyte:** A kilobyte (KB) consists of 1024 bytes.
- c) Megabyte:** A megabyte (MB) consists of 1024 kilobytes, (1024*1024) byte or 1,048,576 bytes) approximately 1,000,000 bytes.
- d) Gigabyte:** A gigabyte (GB) consists of 1024 megabytes, (1024*1024*10240byte) or (1,073,741,824 byte), approximately 1,000,000,000 bytes.
- e) Terabyte:**A terabyte (TB) consists of approximately 1,000,000,000,000 bytes.

Secondary Storage (Backing Storage) is designed to store very large amounts of data for extended periods of time. Secondary storage can have a memory capacity of a gigabyte or more; only small portions of the data are placed in primary storage at any one time. Such as **hard disk drive**, **compact disk(CD)**, **digital versatile disk(DVD)**, **floppy disk**.

❖ Hard disk drive

A- **Internal hard** disk drive such as EIDE, ATA, SATA, SCSI.



B- **External hard disk drive.**



❖ **What is the difference between internal and external hard disks?**

Internal hard disks are located inside your main computer unit, while external hard disks are joined to the main computer unit via a lead that you plug into the back of your computer unit. Some external hard disks will plug into the USB port (connector) located at the back of your computer. Other external hard disks require the installation of a special card within your computer, which allows the connection of the external hard disk to the computer unit.

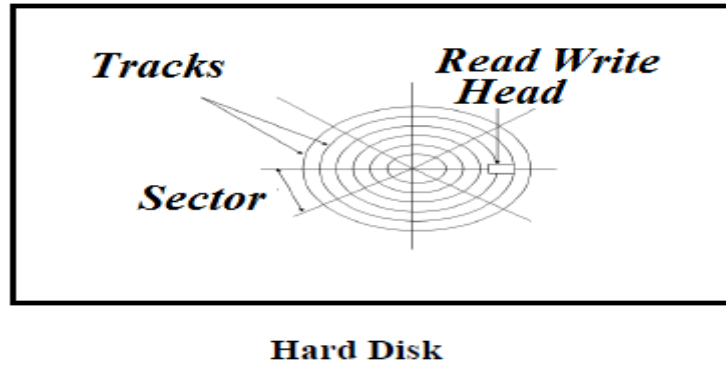
ما الفرق بين الأقراص الصلبة الداخلية والخارجية؟
يوضع القرص الصلب الداخلي داخل وحدة الحاسوب الرئيسية، في حين يتم توصيل القرص الصلب الخارجي بهذه الوحدة عن طريق سلك توصيل تقوم بإدخاله في الجزء الخلفي للوحدة. وبعض الأقراص الصلبة الخارجية يتم توصيلها من خلال المنفذ المسلسل الموجود في الجزء الخلفي من الحاسوب، في حين أن البعض الآخر يحتاج إلى تركيب بطاقة من نوع خاص داخل الحاسوب
لتسمح بتوصيل القرص الصلب بوحدة الحاسوب.

Several basic parameters determine the performance of a given hard disk drive. A seek operation is the movement of the read/write head to the desired track.

1- **Seek Time:** A seeks time is the movement of the read\write head to the desired track. The seek time is the average time for this operation to be performed. Typically, hard disk drives have an average seek time of several milliseconds, depending on the particular drive.

2-**Latency Time:** The latency period is the time takes the desired sector to spin under the head once the head is positioned over the desired track. Latency time depend on the constant rotational speed of the disk.

- The sums of **average seek time** and the **average latency time** is the access time for the disk drive.



The hard disk provides fast access to data. The speed of a hard disk is often quoted as "average access time" speed, measured in milliseconds. **The smaller this number is the faster the disk.**

The hard disk is usually used to back up files you are working on.

تختلف الأقراص الصلبة حسب سرعتها - والتي تتحدد من خلال زمن الوصول إلى البيانات - وهي تقاس بالميلي ثانية. فكلما قل زمن الوصول، زادت سرعة تخزين البيانات على القرص الصلب أو استرجاعها منه. إن السعة التخزينية للأقراص الصلبة في تزايد مستمر، وذلك بالتزامن مع ظهور الأنواع الجديدة من الأقراص. وتقاس هذه السعة بالجيجا بايت. ويساوي الجيجا بايت الواحد 1024 ميغا بايت.

❖ Zip drives 100-200 megabytes.

A Zip disc is like a more recent version of the floppy disk, the main difference being that a single Zip disc can hold up to 250 Mbytes of data. They also offer increased speed compared to the old floppy disk.



هو من وحدات التخزين مثل أقراص الفلوبي إلا أن الفارق أنه يمكن أن يحتوي على 250 ميغا بايت من البيانات. كما أنه يقدم سرعة أكبر في التعامل مع البيانات بالمقارنة بالديسك.

- **Floppy disk (Diskette)**, 1.2-1.4 megabyte. Floppy disks are also known as diskettes. They are very slow compared to hard disks or CD-ROMs and hold relatively small amounts of data (1.44 Mbytes). Sometimes people will backup (copy) important data from their hard disk to floppy disks.

However, as diskettes are notoriously unreliable this is not the best way of backing up valuable data (but is better than nothing).

إن الأقراص المرنة بطيئة جدًا مقارنة بالأقراص الصلبة أو الأقراص المدمجة وتخزن قدرًا صغيرًا نسبيًا من البيانات (1.44 ميجا بايت). وفي بعض الأحيان، يقوم المستخدم بعمل نسخ احتياطية من البيانات المهمة الموجودة على القرص الصلب الذي يستخدمه ويخزنها على أقراص مرنة. ولكن حيث إنها أكثر أقراص التخزين تعرضًا للتلف، فإنها ليست أفضل الطرق لإنشاء نسخ احتياطية من البيانات المهمة.

The floppy is a very cheap mean of backing up 1 mega of text files.
القرص المرن هو وسيلة رخيصة لتخزين واحد ميجا من الملفات النصية

❖ **CD-ROM 700-900 mega:**

Most computers are now supplied with a CD-ROM (Compact Disc - Read Only Memory) drive. CD-ROM discs look exactly like music CDs but contain computer data instead of music. The advantage of a CD-ROM is that it can hold a vast amount of data (equivalent to the storage capacity of over 450 floppy disks). The other big advantage of CD-ROMs is that they are interchangeable.

This means that you can own a range of different CD-ROMs and choose which one to insert into your CD-ROM drive.

معظم الحاسبات الآن مزودة بمشغل الاسطوانات وهي من أنواع ذاكرة القراءة فقط . والسلي دي مثل اسطوانات الموسيقى ولكنها تحتوي على بيانات. وتتميز بأنها تحتوي على كمية كبيرة من البيانات يعني ما يساوي 450 ديسك. وتتميز ايضا بأنه يمكن تغييرها. يعني أنه يمكن أن تمتلك كثير من الأسطوانات وتستخدم الذي ترغب به.



- a) Compact disk read-only memory (CD-ROM) storing devices feature high capacity, low cost.
- b) It has become popular for recorded music as well as information (such as books) a variant is the digital video disk (DVD), used for movies.
- c) Write once, read many (WORM) disk can be written.
- d) The rewritable CD is a less common technology that allows the disk to be written upon and written up to 1.000 times.

❖ DVD 4.7-17 gig.

Short for "Digital Versatile Disc". Similar to CD-ROM drives but allows you to use DVD disks, which contain vastly more information than a traditional CD-ROM disk.

These also transfer the data from the disk to the computer far faster, allowing you to watch movies on your computer screen. A CD-ROM can store 650 MB of data, while a single-layer, single-sided DVD can store 4.7 GB of data. The two-layer DVD standard allows a capacity of 8.5 GB. A double-sided DVD increases the storage capacity to 17 GB (or over 25 times the data storage capacity of a CD-ROM).

إختصار لـ "الأقراص الرقمية". وهي مماثلة لمحركات الأقراص المضغوطة ولكنها تسمح لك باستخدام أقراص دي في دي ، التي تحتوي على معلومات أكثر بكثير من الأقراص المضغوطة التقليدية ، وهذه أيضا تنقل البيانات من القرص إلى جهاز الكمبيوتر أسرع بكثير ، مما يتيح لك بمشاهدة الأفلام على شاشة الكمبيوتر. القرص المدمج يمكنه تخزين 650 ميغا بايت من البيانات ، في حين دي في دي واحد يمكنه تخزين 4.7 جيجابايت من البيانات. الذي في دي ثنائي الطبقة يسمح بتخزين 8.5 جيجابايت. كما يمكن زيادة السعة التخزينية إلى 17 جيجابايت (أو أكثر من 25 مرة من القدرة التخزينية للقرص المضغوط).

❖ Tape backup

A tape backup unit allows for regular backing up of your data. These tapes can store a vast amount of data at a low cost. DAT (Digital Audio Tape) devices are commonly used for backups. The DAT tapes that they use can backup enormous amounts of data (over 4 GBytes per tape). The devices are also fast and reliable.

The tape backup is used to archive personal records that are not currently needed



تتيح لك وحدة أشرطة النسخ الاحتياطي إنشاء نسخ احتياطية من البيانات بشكل منتظم. ويمكن لهذه الأشرطة تخزين كميات كبيرة من البيانات بتكلفة بسيطة. وتستخدم الأشرطة الرقمية DAT (Digital Audio Tape) بكثرة في إنشاء النسخ الاحتياطية. يمكن لأشرطة DAT التي تستخدمها هذه الأجهزة إنشاء نسخ احتياطية لكميات كبيرة من البيانات (أكثر من 4 جيجا بايت لكل شريط). وتتميز هذه الأجهزة بالسرعة والفاعلية. تستخدم أشرطة النسخ الاحتياطي في تخزين السجلات الشخصية التي ليس هناك حاجة إليها.

What is PCMCIA?

Portables by their very nature are very compact and require smaller than standard parts such as hard disks and CD-ROM drives. Many portables are supplied with special adaptor sockets that enable what is called PCMCIA compatible hardware to be connected to them.

PCMCIA components tend to be more expensive than standard computer parts that are designed for more bulky desktop computers.

إن الحواسيب المحمولة بطبيعتها مضغوطة جدا وتحتاج إلى مكونات أصغر حجماً من المكونات القياسية للحاسوب العادي مثل الأقراص الصلبة ومشغلات الأقراص المدمجة. وتشتمل العديد من الحواسيب المحمولة على مقابس خاصة في الجزء الخلفي منها والتي تستخدم في توصيلها بالأجهزة المتوافقة مع ما يعرف بـ PCMCIA. إن الأجهزة المتوافقة مع PCMCIA تكون عادة أعلى من الأجهزة القياسية المصممة للعمل مع الحواسيب المكتبية الأكبر حجماً.

The differences between the PRIMARY and Secondary storage

Primary storage has much less capacity than secondary storage and is faster and more expensive per byte stored, primary storage is located much closer to the CPU than secondary storage. Sequential _access

secondary storage media such as magnetic tape is much slower and less expensive than direct access media (e.g. hard drives, optical media).

Why format a disk?

Originally, when you purchased a pack of floppy disks (diskettes), you had to format them prior to use. Today, most floppy disks are supplied pre-formatted.

Formatting a disk is like putting lines on a blank sheet of paper. Formatting allows the operating system (i.e. Windows) to read information stored on the disk and to store information on the disk. In the same way, the manufacturer formatted your hard disk. You should be very careful about formatting a disk, as any data on the disk will be lost after re-formatting.

عندما تشتري علبة من الأقراص المرنة تحتاج إلى تهيئتها قبل الاستخدام. هذه الأيام معظم الديسكات التي يتم شراؤها تكون سابقة التهيئة (الفورمات). عملية الفورمات للديسك مثل وضع خطوط على ورقة بيضاء. الفورمات يسمح لنظام التشغيل بقراءة البيانات الموجودة على الديسك وكذلك تسجيل البيانات عليه. يجب أن تكون حريصا جدا عند عملية التهيئة للأقراص حيث أن أي بيانات على القرص سوف تفقد عند إعادة التهيئة.

Summary

- ❖ Unformatted disks need operating system data written to them before they can store files.

الديسكات التي لم يسبق تهيئتها تحتاج إلى نظام تشغيل يقوم بإعدادها حتي يمكن ادخال بيانات عليها.

- ❖ A used floppy disk can be reformatted.

الديسك المستخدم يمكن إعادة تهيئته.

- ❖ A pre-formatted disk can be used immediately to store data.

الديسكات التي سبق تهيئتها يمكن ان تستخدم مباشرة لادخال البيانات عليها.

- ❖ Full disk formatting completely erases the previous contents of a disk.

التهيئة الكاملة للأقراص تمسح كل البيانات الموجودة عليها مسبقا.

- ❖ All disks need to be formatted before they can store files.

كل الديسكات تحتاج إلى تهيئة حتي يمكن تخزين بيانات عليها.

Why do you need to back up your computer?

The most important thing that you store on your computer is information. Often the contents of a hard disk can represent years of work. If the hard disk stops working one day, you could lose all those years of work. For this reason, it is vital (ضروري) that you take regular backups of the information that is stored on the computer.

In large organizations, this backup procedure is normally performed automatically by your computer support team, where the data is normally held on a centralized, networked computer.

إن أهم شيء تقوم بتخزينه على الحاسوب هي المعلومات. فعادةً ما تمثل محتويات القرص الصلب خلاصة سنوات من العمل. فإذا توقف القرص الصلب عن العمل، من الممكن أن تفقد كل هذا المجهود. ولذلك، من الواجب عليك إنشاء نسخ احتياطية من المعلومات المخزنة على الحاسوب. وفي الشركات الكبيرة، حيث يتم تخزين البيانات على حاسوب مركزي موصل بشبكة الشركة، يقوم فريق الدعم الخاص بالحاسوب بعمل النسخ الاحتياطية.

In smaller organizations, it is often up to the individual to organize some sort of data backup. If nothing else is available, copy your files to a floppy disk and make sure that these backup disks are stored away from the computer, ideally off-site. If there is a fire and your office burns down if your backup disks are stored next to the computer they too will be incinerated. Back up is copying your computer files to a separate storage device. Back up is keeping copies of data in two different locations.

وفي الشركات الصغيرة، عادة ما يترك الأمر للشخص بالنسبة لتنظيم عملية النسخ الاحتياطي للبيانات. وإذا لم تكن هناك طرق أخرى محددة للنسخ الاحتياطي، قم بنسخ هذه الملفات على أقراص مرنة وتوضع هذه الأقراص في مكان بعيد عن الحاسوب، وذلك تحسباً لحدوث أية مشكلات في المكان الذي يوجد به الحاسوب.

Part 2 of Chapter 2

Main Parts of a Personal Computer

The basic computer parts are the motherboard, processor, Ram, hard drive and power supply. The main parts are parts that must be in to make the computer work properly; the next sections explain some of the most important parts.

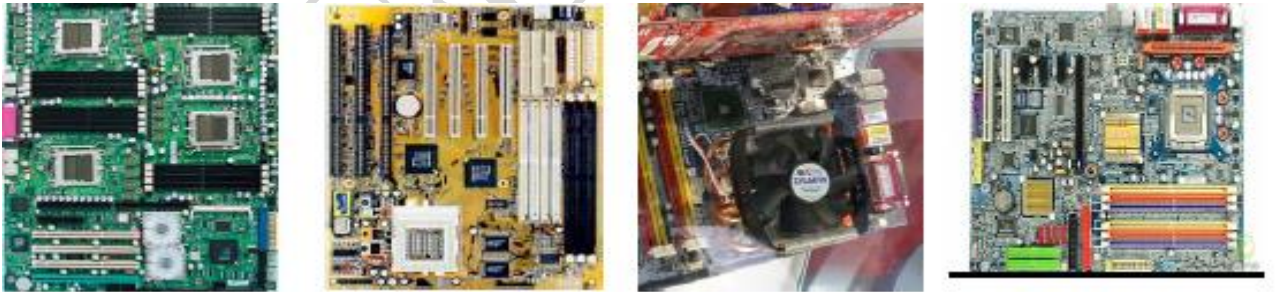
1- The system unit (case): The "system unit" is the name given to the main PC box that houses the various elements that go together to make up the PC. For instance within the system unit is the computer system's motherboard, which contains all the main components, such as the CPU. The system unit also houses items such as the hard disk, the floppy disk, and CD-ROM drives etc. System units come in two basic varieties, the tower version, as illustrated, or a desktop version, which is designed to sit on your desk with your monitor on top of the system unit.



يطلق هذا الاسم على الصندوق الرئيسي الخاص بالحاسوب الشخصي والذي يحتوي على المكونات المختلفة التي يتألف منها جهاز الحاسوب. فتوجد بهذه الوحدة - على سبيل المثال - اللوحة الأم (لوحة النظام) والتي تشتمل على كل الأجزاء الرئيسية، مثل المعالج. ويوجد بها أيضاً مكونات أخرى مثل القرص الصلب ومشغلات الأقراص المرنة والمدمجة وهكذا. وهناك نوعان لهذه الوحدة وهما الوحدة الرأسية - والتي تظهر في الشكل المقابل - والوحدة الأفقية، التي تم تصميمها لكي توضع على المكتب الذي تعمل عليه وتوضع الشاشة فوقها.

2- The System (Mother) Board

The system (mother) board is contained within your system unit and all the vital computer systems plug directly into the system board. The CPU is normally housed on your system board along with all the other electronic components. Other items as the hard disk are attached to the system board, either directly or via -cables. These boards are getting smaller and smaller as the components become more integrated. If you open up a modern system, you will find that it is mainly full of air.



توجد اللوحة الأم (لوحة النظام) داخل وحدة النظام ويتم توصيل كل مكونات الحاسوب الأساسية بشكل مباشر بها. وتوجد وحدة المعالجة المركزية في لوحة النظام إلى جانب المكونات الإلكترونية الأخرى. ويتم توصيل مكونات أخرى مثل القرص الصلب بهذه اللوحة بشكل مباشر أو من خلال كابلات. إن هذه اللوحات يقل حجمها يوماً بعد يوم لأن المكونات تصبح أكثر دمجاً. فإذا قمت بفتح أية وحدة حاسوب حديثة، فلن تجد غير مكونات صغيرة جداً.

Some of the prime functions of a computer motherboard are as follows:

- ❖ The motherboard acts as the central backbone of a computer on which other modular parts are installed such as the CPU, RAM, and hard disks.

تعمل اللوحة الأم باعتبارها العمود الفقري المركزي لجهاز الكمبيوتر الذي يتم تثبيت الأجزاء النمطية الأخرى عليه مثل وحدة المعالجة المركزية وذاكرة الوصول العشوائي والأقراص الصلبة

- ❖ The motherboard also acts as the platform on which various expansion slots are available to install other devices/interfaces.

تعمل اللوحة الأم أيضًا كنظام أساسي تتوفر عليه فتحات التوسعة المختلفة لتثبيت الأجهزة / الواجهات الأخرى.

- ❖ The motherboard is also responsible to distribute power to the various components of the computer.

اللوحة الأم مسؤولة أيضًا عن توزيع الطاقة على مكونات الكمبيوتر المختلفة.

- ❖ They are also used in the coordination of the various devices in the computer and maintain an interface among them.

كما أنها تستخدم في تنسيق الأجهزة المختلفة في الكمبيوتر والحفاظ على واجهة فيما بينها

2.1 Chipset

The chipset is the "glue" that connects the microprocessor to the rest of the motherboard and therefore to the rest of the computer. On a PC, it consists of two basic parts -the **Northbridge** and the **Southbridge**. All of the various components of the computer communicate with the CPU through the chipset.

Why there is a need to have a chipset on the motherboard

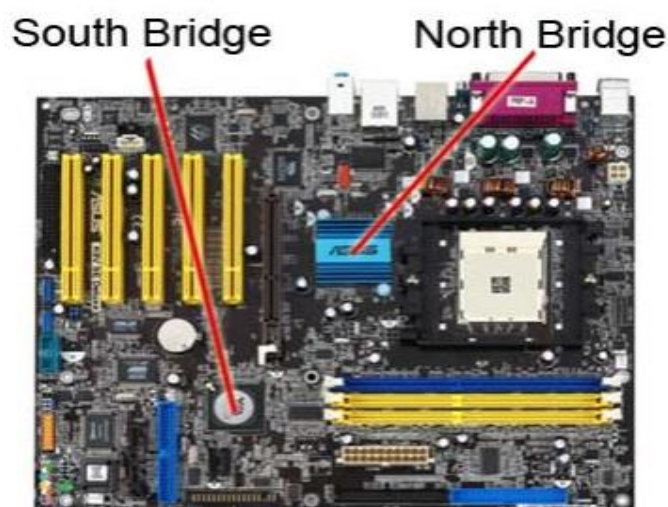
The chipset consists of two parts, they called the **Northbridge** and the **Southbridge**, as we mentioned above.

The Northbridge chip is used to connect all the major components of the motherboard with each other. It is used to connect the CPU to the memory, cache, PCI express bus etc.

(Peripheral Component Interconnect (PCI) slots are such an integral part of a computer's architecture that most people take them for granted. For years, PCI has been a versatile, functional way to connect sound, video and network cards to a motherboard)

The Southbridge chip is used to handle all the communication between the various input and output devices. Both the Northbridge and the southbridge together form a chipset for a motherboard. The southbridge also makes use of the bus as a communication channel.

Note that the northbridge considers as the fast part of the chipset while the southbridge is the slow end.



The main reason that the chipset is divided into two separate units is that it is difficult to integrate the various components on a single chip. Moreover, the accessing/transfer speed of instructions/commands/data between the processor and other components (some of the components processes/interprets slowly) is different, so Northbridge provides fast access as compared to Southbridge.

Northbridge is the part of the chipset which communicates with the processor. As its interaction is with the processor, so it is fast. It acts as a medium channel between processor and memory.

Southbridge is slower than the Northbridge and it connects and controls computer Input-output functions. It controls Universal Serial Bus – USB, the system Basic Input-output System – BIOS, serial ports etc.



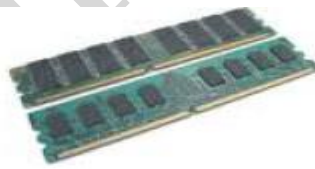
3- The CPU

The CPU (Central Processing Unit) is one of the most important components within your computer(as we mentioned before). It determines how fast your computer will run and is measured by its MHz speed. Sometimes referred to the central processor, but more commonly called processor. Usually, each motherboard will support a specific type of CPU.

The CPU performs all the computer's operations. The faster the processor is, the quicker it can perform and the faster your computer runs.

4-Memory (RAM)

The RAM (Random Access Memory) within your computer is where the operating system is loaded to when you switch on your computer and also where your applications are copied to when you start an application, such as a word processor or database program. When you create data, (e.g. letters and pictures), these are initially created and held in RAM and then copied to disk when you save the data. As a rule of thumb, the more RAM you have installed on your computer the better.



إن ذاكرة الوصول العشوائي (RAM) التي توجد داخل أي حاسوب هي المكان الذي يتم تحميل نظام التشغيل إليه عندما يتم بدء تشغيل الحاسوب وأيضا يتم إليه نسخ البرامج التطبيقية وتحميلها، على سبيل المثال: برامج قواعد البيانات أو معالجة النصوص. فعندما تقوم بإنشاء بيانات (على سبيل المثال، أحرف وصور)، يتم إنشاؤها في ذاكرة الوصول العشوائي ثم نسخها إلى قرص عند قيامك بحفظ البيانات.

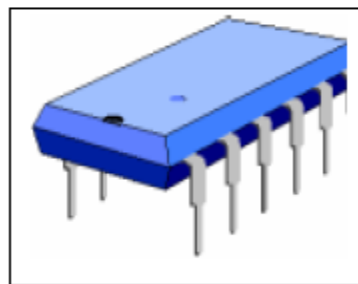
- The operating system is loaded into RAM when the PC is switched on.
- The contents of RAM change constantly as the PC is used.
- The contents of this disappear when the PC is switched off.

5- What is ROM?

Read Only Memory (ROM) as the name suggests is a special type of memory chip that holds software that can be read but not written to. A good example is a ROM-BIOS chip, which contains read-only software. Often network cards and video cards also contain ROM chips.

- **The ROM-BIOS**

The 'Read Only Memory Basic Input Output System' chip is a chip located on the computer's system (mother) board, which contains software. This software performs a variety of tasks. When you first switch on the computer, the ROM-BIOS software performs a self-diagnostic to check that the computer is working OK. This software then loads your operating system from the disk into the RAM.



إن ذاكرة القراءة فقط المخصصة لنظام الإدخال والإخراج الأساسي هي شريحة خاصة يتم تركيبها في نظام الحاسوب على اللوحة الأم. وهي تحتوي على البرامج اللازمة لجعل الحاسوب يعمل مع نظام التشغيل، مثلاً: هي مسؤولة عن نسخ نظام التشغيل الذي تستخدمه إلى ذاكرة الوصول العشوائي عند قيامك ببدء تشغيل الحاسوب.

Summary

- The contents of this(ROM) are fixed during the manufacture of the pc.
- You cannot write to this type of memory.
- It cannot be modified by the user.
- This type of memory is permanent(دائم) and unchanging.

What is flash BIOS ?

Most modern computers are actually supplied with a flash BIOS rather than a ROM-BIOS. This chip contains exactly the same type of inbuilt software but has the advantage that the software on the chip can be upgraded. This upgrade is achieved by simply running a small program supplied by the computer manufacturer.

6- Hard (Fixed) Disk

Hard disks are the main, large data storage area within your computer. Hard disks are used to store your operating system, your application programs (your word processor, games etc) and your data. They are much faster than CD-ROMs and floppy disks and can hold much more data.(mentioned to this device before)

تعد الأقراص الصلبة مساحة التخزين الرئيسية والكبيرة الموجودة داخل الحاسوب الخاص بك. وتستخدم الأقراص الصلبة في تخزين نظام التشغيل والبرامج التي تستخدمها (على سبيل المثال، برنامج معالجة النصوص والألعاب وهكذا) والبيانات الخاصة بك. وهي أسرع بكثير من الأقراص المدمجة والمرنة ويمكنها تخزين قدر أكبر بكثير من البيانات.

7 -The Monitor

An Output device. The monitor is the TV type screen that you view your programs on. They are supplied in different sizes, common sizes range from 15" to 21" screens. You should be aware that poor quality or badly maintained monitors could harm your eyesight.

The VDU (Visual Display Unit) is the computer screen used for outputting information in an understandable format for humans. Remember that at the end of the day the computer works in binary code (a series of on/off impulses). It is hard to realize that the original electronic computers did not have a screen.

إن شاشة العرض المرئية هي شاشة من النوع الذي يستخدم مع أجهزة التلفاز والتي تعرض من خلالها البرامج. وهناك أحجام كثيرة للشاشات تتراوح ما بين 15 إلى 21 بوصة. ويجب أن تعرف أن الشاشات ذات الجودة المتدنية أو التي لا تتم صيانتها بشكل جيد يمكن أن تضر عينيك.

7.1 - Flat screen monitors

Traditional computer monitors are based on the same sort of technology that is used within a television screen.

More recently, flat screen computer monitors have become available. These take up a lot less room on a desk and use less energy than the traditional, more bulky monitors.

تستند شاشات الكمبيوتر التقليدية على نفس النوع من التكنولوجيا التي تستخدم داخل شاشة التلفزيون. وفي الآونة الأخيرة، أصبحت شاشات الكمبيوتر بشاشة مسطحة المتاحة. هذه تأخذ مساحة أقل كثيراً على مكتب وتستخدم طاقة أقل من أجهزة العرض التقليدية، الأكبر حجماً

- VDU(visual display unit)
- CRT stands for (cathode ray tube).
- LCD stands for (liquid crystal display).

7.1.1 The Cathode Ray Tube(CRT):

The CRT used in video monitor is shown in the figure below

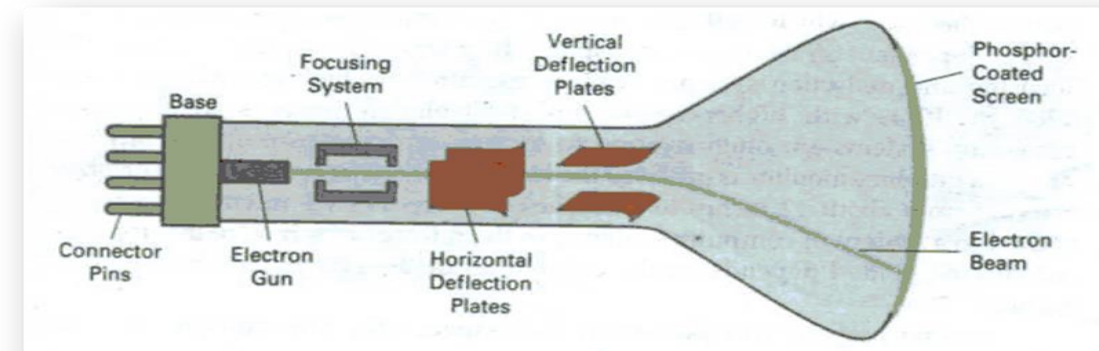


Figure (1-1):- Electrostatic deflection of the electron beam in a CRT

Consist of an electron gun that contains a cathode that when heated emits a beam of negatively charged electrons towards a positively charged phosphor-coated screen.

The electron beam passes through the focusing and deflection system, which consist of an electrostatic field.

The focusing system concentrates the beam so that by the time the electrons reach the screen. They have converged to a small dot.

The deflection system which consists of two pairs of deflection plates (horizontal and vertical) directs the electron beam to any point on the screen. When the electron beam strikes the screen, the phosphor emits a spot of visible light who intensity depends on the

number of electrons on the beam. The light on the screen starts to fade as soon as the beam moves to another location.

Two types of CRT of refresh CRTs are available raster scan and random vector

7.1.2 Flat-panel display (LCD):

Compared to CRT's, flat panel display are much thinner, weightless, consuming less power, and do not emit radiation. Thus they are better for portable computers. There are several types of flat-panel monitors, but the most common is the Liquid Crystal Display (LCD).

The disadvantage of LCD is that their image can be difficult to see in bright light. For this reason, laptop computers users often look for shady places to sit when working outdoors or near windows. A bigger disadvantage of LCD monitors is their limited viewing angle. With most CRT, you can see the image clearly even when standing at an angle to the screen. In LCD monitors, the viewing angles shrink; as you increase your angle to the screen, the image becomes fuzzy quickly.

In many older flat panel systems, the user must face the screen nearly straight on to the image clearly. Technological improvement has extended the viewing angles of flat-panel monitors while causing their price to increase.

7.2 Screen clarity

The screen clarity depends on three qualities: resolution, dot pitch, refresh rate.

- **Resolution:** the clarity or sharpness of the display screen is called resolution, the more pixels per square inch, and the better of resolution. A high-resolution graphic has more dots(pixels) per a specific area, while a low-resolution graphic has a lower dots or pixels. The resolution applies to both graphics designed for print and graphics created for the display system. A screen with 640×480 pixels multiplied together equals 307200 pixels. the screen will be less clear and sharp than a screen with 800×600 (equals 480,000) or 1024×768 (equals 786,432)pixels.

Note:- Resolution depends on the type of phosphor, the intensity to be displayed and the focusing and deflection systems used in the CRT.

- **Dot pitch:** is the amount of space between the centers of adjacent pixels, the closer the dots, the crisper the image. For crisp images, dot pitch should less than 0.31 millimeter. On CRTs, the dot pitch is typically from 0.28 to 0.51mm, while large presentation monitors may go up to 1 mm. On LCD monitors, dot pitch is typically from 0.16 to 0.29mm

- **Refresh rates:-**refresh rate is the number of times per second that the pixels recharged so their glow remains bright. in dual-scan screens, the top and the bottoms of the screens are refreshed independently at twice the rate

of single scan screen, producing more clarity and richer colors, in general, displays are refreshed 45 to 100 times per second.

7.3 - Screen size

You should be aware that often if you specify a screen of a certain size, say a 17-inch screen, that this is the size measured diagonally, not horizontally across the screen. If you are upgrading you should also ask for the "visible viewing area" of the screen.

8- Sound cards and speakers

Many computers are now supplied with sound cards and speakers that mean that when you run 'multi-media' programs, you can listen to sounds that are played back via your computer. If you have a microphone and suitable software, you can also record sounds. You can even purchase special software that will allow you to talk to your computer and get the computer to type the words you have spoken on your screen. In time, this type of software may replace the keyboard.



معظم الحاسبات الحديثة مزودة بكروت صوت وسماعات وهذا يعني أنه عند تشغيل برامج الملتيميديا يمكنك سماع الأصوات التي تصدر من الحاسب. وإذا كان لديك ميكروفون وبرنامج مناسب فإنه يمكنك تسجيل الصوت وإدخاله للحاسب. وأيضاً يمكنك شراء برامج خاصة تمكنك من التحدث للحاسب فيقوم بكتابته ما تقوله على الشاشة. في الوقت الحالي يمكن لهذا البرنامج ان يحل محل لوحة المفاتيح.

9- Serial Port

The serial port is a socket located at the back of your computer, which enables you to connect items to the computer, such as a modem. They are commonly labeled as COM1 or COM2.



Parallel Port

The parallel port is a socket located at the back of your computer, which enables you to connect items to the computer, such as a printer. It is commonly labeled as LPT1 or LPT2.

Universal Serial Bus (USB)

- The Universal Serial Bus is a relatively new item on the PC. You will see one or more USB sockets at the back of the system unit, allowing you to plug in devices designed for the USB. These devices include printers, scanners, and digital cameras.

